Bureau
International des
Poids et
Mesures

Le système international d’unités

# The International System of Units

**9 édition 2003**

Annexe 2

**Annex 2**

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

## Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2

**Part 2**

Iodine ()

## Iodine ()

CCL-CCTF Frequency Standards Working Group

Bureau International
des Poids et Mesures

# Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
# Part 2: Iodine ()

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Appendix 2

# 1. CIPM recommended values

The values $f = 468\ 218\ 332.4$ MHz
$\lambda = 640\ 283\ 468.7$ fm

with a relative standard uncertainty of $4.5 \times 10^{-10}$ apply to the radiation of a He-Ne laser stabilized with an internal iodine cell having a cold-finger temperature of $(16 \pm 1)$ °C[1] and a frequency modulation width, peak-to-peak, of $(6 \pm 1)$ MHz.

[1] For the specification of operating conditions, such as temperature, modulation width and laser power, the symbols ± refer to a tolerance, not an uncertainty.

# 2. Source data

Adopted value

$f = 468\ 218\ 332.4\ (2)$ MHz
$u_c/y = 4.5 \times 10^{-10}$

for which:
$\lambda = 640\ 283\ 468.7\ (3)$ fm
$u_c/y = 4.5 \times 10^{-10}$

calculated from

| $f$/ kHz | $u_c/y$ | source data |
|---|---|---|
| 468 218 332 419 | $1.0 \times 10^{-10}$ | Section 2.1 |
| 468 218 332 310 | $1.2 \times 10^{-10}$ | Section 2.2 |
| 468 218 332 069 | $4.6 \times 10^{-10}$ | Section 2.3 |
| Weighted mean | $f = 468\ 218\ 332\ 366$ kHz | |

Given the small number of determinations, the CCL considered it prudent to assume a relative standard uncertainty of $4.5 \times 10^{-10}$.

Source data

## 2.1.

Bennet and Mills-Baker [1] give $\lambda_{a9} = 640.283\ 468\ 6$ nm.

From this paper the ratio $f_{a9}/f_i$ is calculated as

$f_{a9}/f_i = 0.988\ 611\ 184\ 191$
$u_c/y = 1 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_i = 473\ 612\ 214\ 712$ kHz
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a9} = 468\ 218\ 332\ 434$ kHz
$u_c/y = 1.0 \times 10^{-10}$

at a cold-finger temperature of 14.3 °C (iodine pressure = 16 Pa) and a modulation width of 7 MHz. For a reference temperature of 16 °C (iodine pressure = 18.9 Pa) and a modulation width of 6 MHz, corrections of −23 kHz and + 8 kHz has to be applied to this value assuming a pressure dependence of −7.8 kHzPa$^{-1}$ and a modulation dependence of −7.6 kHzMHz$^{-1}$, similar to that reported in [2], giving

$f_{a9} = 468\ 218\ 332\ 419$ kHz

$u_c/y = 1.0 \times 10^{-10}$.

## 2.2.

Zhao et al. [2], [3] give $\lambda_{a9} = 640.283\ 468\ 8\ \text{nm}$
$3 \times (u_c/y) = 1.1 \times 10^{-9}$.

Bönsch [4] gives $\lambda_i/\lambda_{a9} = 0.988\ 611\ 183\ 8$
$u_c/y = 12 \times 10^{-11}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_i = 473\ 612\ 214\ 712\ \text{kHz}$
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a9} = 468\ 218\ 332\ 277\ \text{kHz}$
$u_c/y = 1.2 \times 10^{-10}$

at a cold-finger temperature of 18 °C (iodine pressure = 22.6 Pa) and a modulation width of 6.5 MHz. For a reference temperature of 16 °C (iodine pressure = 18.9 Pa) and a modulation width of 6 MHz, corrections of + 29 kHz and + 4 kHz have to be applied to this value knowing a pressure dependence of −7.8 kHzPa$^{-1}$ and a modulation dependence of −7.6 kHzMHz$^{-1}$, giving

$f_{a9} = 468\ 218\ 332\ 310\ \text{kHz}$
$u_c/y = 1.2 \times 10^{-10}$.

## 2.3.

Reference [5], [6] give $\lambda_{a9}$ (17 °C)/$\lambda_e$ (20 °C) = 1.011 520 341 04
$u_c/y = 4.6 \times 10^{-10}$.

Using the recommended value of the absorbing molecule $^{127}I_2$, $a_{16}$ or f component, R(127) 11-5 transition (see iodine at $\lambda \approx 633$ nm and frequency differences listed in corresponding Table 1, see p. 8) one obtains

$f_e = 473\ 612\ 366\ 967\ \text{kHz}$
$u_c/y = 2.2 \times 10^{-11}$,

one calculates

$f_{a9} = 468\ 218\ 332\ 055\ \text{kHz}$
$u_c/y = 4.6 \times 10^{-10}$

at a cold-finger temperature of 17 °C (iodine pressure = 20.7 Pa). For a reference temperature of 16 °C (iodine pressure = 18.9 Pa), a correction of + 14 kHz has to be applied to this value, assuming a pressure dependence of −7.8 kHzPa$^{-1}$ similar to that reported in [2], giving

$f_{a9} = 468\ 218\ 332\ 069\ \text{kHz}$

$u_c/y = 4.6 \times 10^{-10}$.

# 3. Absolute frequency of the other transitions related to those adopted as recommended and frequency intervals between transitions and hyperfine components

These tables replace those published in BIPM Com. Cons. Long., 2001, **10**, 188 and *Metrologia*, 2003, **40**, 128.

The notation for the transitions and the components is that used in the source references. The values adopted for the frequency intervals are the weighted means of the values given in the references.

For the uncertainties, account has been taken of:

- the uncertainties given by the authors;
- the spread in the different determinations of a single component;
- the effect of any perturbing components;
- the difference between the calculated and the measured values.

In the tables, uc represents the estimated combined standard uncertainty ($1\sigma$).

All transitions in molecular iodine refer to the B-X system.

**Table 1**

| $\lambda \approx 640$ nm $^{127}I_2$ P(10) 8-5 | | | | | |
|---|---|---|---|---|---|
| $a_n$ | $[f(a_n)-f(a_9)]$/MHz | $u_c$/ MHz | $a_n$ | $[f(a_n)-f(a_9)]$/MHz | $u_c$/ MHz |
| $a_1$ | -495.4 | 0.4 | $a_9$ | 0 | — |
| $a_2$ | -241.5 | 0.7 | $a_{10}$ | 77.84 | 0.03 |
| $a_3$ | -233.0 | 0.4 | $a_{11}$ | 186.22 | 0.07 |
| $a_4$ | -177.8 | 1.3 | $a_{12}$ | 199.51 | 0.07 |
| $a_5$ | -175.2 | 0.6 | $a_{13}$ | 256.6 | 0.2 |
| $a_6$ | -130.8 | 0.1 | $a_{14}$ | 272.75 | 0.07 |
| $a_7$ | -82.45 | 0.03 | $a_{15}$ | 374.0 | 0.2 |
| $a_8$ | -61.85 | 0.14 | | | |
| Frequency referenced to | $a_9$, P(10) 8-5, $^{127}I_2$: $f = 468\,218\,332.4$ MHz [7] | | | | |

Ref. [8], [9], [10], [11], [12], [13], [14], [15]

**Table 2**

| $\lambda \approx 640$ nm $^{127}I_2$ R(16) 8-5 | | |
|---|---|---|
| $b_n$ | $[f(b_n)-f(a_9)]$/MHz | $u_c$/ MHz |
| $b_1$ | 62.834 | 0.01 |

| | | |
|---|---|---|
| $b_2$ | 329.8 | 0.2 |
| $b_3$ | 335.99 | 0.02 |

Frequency referenced to $a_9$, P(10) 8-5, $^{127}I_2$: $f = 468\ 218\ 332.4$ MHz [7]

Ref. [8], [9], [10], [11], [12], [13], [14], [15]

## Absorbing molecule $^{127}I_2$, $a_9$ component, P(10) 8-5 transition[1]

[1] All transitions in $I_2$ refer to the $B^3\Pi\ 0_u^+ - X^1\ \Sigma_g^+$ system.

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